Assignment 7 Solution

```
install.packages("lme4")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.3     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.4.4     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.0
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lme4)
```

```
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:dplyr':

    expand, pack, unpack

Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select

The following objects is masked from 'package:purrr':

    map

library(MASS), rotation = "vertical"
library(lme4)
```

```
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
my_url <- "http://datalab.tlu.ee/images.tlu/ms/ff-octave.csv"
self_octave <- read_csv(my_url)
```

```
Rows: 20 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): subject
dbl (7): repeat, target, tempo, self_octave, self_note, self_rhythm, self_...

Warning: The `x` argument of `as_tibble.matrix()` must consist of vectors, but
`...1` is not.
Warning: Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
self_octave
```

```
# A tibble: 20 × 8
   subject repeat target self_octave self_note self_rhythm self_tempo
   <chr>   <dbl>  <dbl>      <dbl>     <dbl>     <dbl>     <dbl>     <dbl>
 1 S       Control    4          3         3         3        14        13
 2 S       Control    5          3         3        14        13        13
 3 S       Control    6          3         3        13        13        13
 4 S       Control    7          3         3         3        13        12
 5 S       Control    8          3         3        13        13        12
 6 S       Control    9          4         4        17        16        16
 7 S       Control   10          4         4        17        16        16
 8 S       Control   11          4         4        16        16        16
 9 S       Control   12          4         4        16        16        16
10 S       Control   13          3         3        14        13        13
```

Just subject is a name of the experimental produce three measurements, namely the self-octave scores at the three times. These are the "repeated measures". Additionally, the repeated measures analysis will be necessary because the three scores on the same subject may not be correlated.

```
self_octave %>% pivot_longer(cols=self_octave:self_tempo)
```

```
# A tibble: 60 × 4
   subject repeat target name      value
   <chr>   <dbl>  <dbl> <chr>       <dbl>
 1 S       Control    4 self_octave     3
 2 S       Control    4 self_note       3
 3 S       Control    4 self_rhythm     3
 4 S       Control    4 self_tempo     13
 5 S       Control    5 self_octave     3
 6 S       Control    5 self_note       3
 7 S       Control    5 self_rhythm    14
 8 S       Control    5 self_tempo     13
 9 S       Control    6 self_octave     3
10 S       Control    6 self_note       3
11 S       Control    6 self_rhythm    13
12 S       Control    6 self_tempo     13
13 S       Control    7 self_octave     3
14 S       Control    7 self_note       3
15 S       Control    7 self_rhythm     3
16 S       Control    7 self_tempo     12
17 S       Control    8 self_octave     3
18 S       Control    8 self_note       3
19 S       Control    8 self_rhythm    13
20 S       Control    8 self_tempo     12
21 S       Control    9 self_octave     4
22 S       Control    9 self_note       4
23 S       Control    9 self_rhythm    17
24 S       Control    9 self_tempo     16
25 S       Control   10 self_octave     4
26 S       Control   10 self_note       4
27 S       Control   10 self_rhythm    17
28 S       Control   10 self_tempo     16
29 S       Control   11 self_octave     4
30 S       Control   11 self_note       4
31 S       Control   11 self_rhythm    16
32 S       Control   11 self_tempo     16
33 S       Control   12 self_octave     4
34 S       Control   12 self_note       4
35 S       Control   12 self_rhythm    16
36 S       Control   12 self_tempo     16
37 S       Control   13 self_octave     3
38 S       Control   13 self_note       3
39 S       Control   13 self_rhythm    14
40 S       Control   13 self_tempo     13
41 S       Control    4 self_octave     3
42 S       Control    4 self_note       3
43 S       Control    4 self_rhythm    14
44 S       Control    4 self_tempo     13
45 S       Control    5 self_octave     3
46 S       Control    5 self_note       3
47 S       Control    5 self_rhythm    13
48 S       Control    5 self_tempo     13
49 S       Control    6 self_octave     3
50 S       Control    6 self_note       3
51 S       Control    6 self_rhythm     3
52 S       Control    6 self_tempo     13
53 S       Control    7 self_octave     3
54 S       Control    7 self_note       3
55 S       Control    7 self_rhythm    13
56 S       Control    7 self_tempo     12
57 S       Control    8 self_octave     3
58 S       Control    8 self_note       3
59 S       Control    8 self_rhythm    13
60 S       Control    8 self_tempo     12
```

```
self_octave %>%
  ggplot(aes(x=repeat, y=self_octave))
  + ms:dot(ci) + stepwise
```

```
ggplot(response, x = DV)
```

```
  ant ant ant
1   3   3   3
2   3   3   3
3   3   3   3
4   3   3   3
5   3   3   3
6   3   3   3
7   3   3   3
8   3   3   3
9   3   3   3
10  3   3   3
11  3   3   3
12  3   3   3
13  3   3   3
14  3   3   3
15  3   3   3
16  3   3   3
17  3   3   3
18  3   3   3
19  3   3   3
20  3   3   3
Index: c(ant, ant, ant)
Index: c(3, 3, 3)
```

```
Index <- c(response)
Index[ant] <- data.frame(Index = factor(Index))
```

```
self_octave.1 <- lm(response ~ regime, data = self_octave)
```

```
Error in `contrasts<-.factor(*tmp*, 2, value = c(ant, ant, ant))`: 'contrasts' can be assigned only to factors with 2 or more levels
```

Take by Mixed Interaction linear model, see: Elliott, 2 to describe the repeated time structure, and run a repeated-measures ANOVA with time as the within-subject factor:

`aov()` and `Manova()` are different functions from different packages, even though their names look almost the same. The lowercase `aov()` is the base R function for a standard ANOVA, which does not have an error term to express variables and score to test whether groups differ across them. The capitalized `Manova()` from the `car` package, is more flexible and is especially useful for repeated measures ANOVAS because it lets us specify the within-subject structure using `idata` and `idesign`. Unlike `aov()`, `Manova()` requires a fitted multivariate model (`mlm`) and uses these extra arguments to run the correct repeated-measures analysis automatically. At the same outcome measure, `aov()` and `Manova()` will usually give identical results when the design is simple, but only `Manova()` understands repeated-measures. The `Anova()` function does not handle that step in the same direct way, while `Manova()` does.

```
summary(self_octave.1)
```

```
Type II Repeated Measures MANOVA Tests:
---------------------------------------

Term: (Intercept)

  Response transformation matrix:
  (Intercept)
ant           1
ant           1
ant           1

Sum of squares and products for the hypothesis:
            (Intercept)
(Intercept)  1.08e+04

Multivariate Tests: (Intercept)
                 df test stat approx F num df den df Pr(>F)
Pillai           1  0.99000 1080.000      1     19 < 2.2e-16
Wilks            1  0.01000 1080.000      1     19 < 2.2e-16
Hotelling-Lawley 1 53.89737 1080.000      1     19 < 2.2e-16
Roy              1 53.89737 1080.000      1     19 < 2.2e-16

Sphericity tests:
Error SS and SP exceed square if num DF dev DF   Pr(>F)
Mauchly                   1                 0 NaN     0 NaNe+00
Greenhouse-Geisser        1                 0 NaN     0 NaNe+00
Huynh-Feldt               1                 0 NaN     0 NaNe+00

Univariate Type II Repeated-Measures ANOVA Assuming Sphericity:
                SS num Df Error SS den Df F value    Pr(>F)
(Intercept)  1080      1     0     19     Inf 2.6e-204
regime          0      2     0     38     NaN       NaN
regime:Sex      0      2     0     38     NaN       NaN
regime:name     0      2     0     38     NaN       NaN

Mauchly Tests for Sphericity:
                     Test statistic p-value
regime                           NA      NA
regime:Sex       0.7426 0.4117117
regime:name      0.7426 0.4117117


Greenhouse-Geisser and Huynh-Feldt Corrections
 for Departure from Sphericity:

                 GG eps Pr(>F[GG])
regime:Sex       0.8091  0.7286147
regime:name      0.8091  0.7286147

Error: Error(regime:name)
                 Df Sum Sq Mean Sq F value Pr(>F)
regime:name       2      0       0       NaN    NA
Residuals        38      0       0
Warning message:
In summary.lm(.Y, ...) :
  summary.lm() produced output by "regime". You can override using the
  "print" argument.
```

```
self_octave.1 <- lm(
  cbind(self_octave, self_note, self_rhythm, self_tempo)
  ~ regime, data = self_octave)
```

```
Error in `model.frame.default(formula = cbind(self_octave, self_note,
  self_rhythm, self_tempo) ~ regime, data = self_octave,
  drop.unused.levels = TRUE)`: 'data' must be of a vector type, was 'NULL'
```

```
summary(self_octave.1, multivariate = TRUE)
```

```
Error in `model.frame.default(formula = cbind(self_octave, self_note,
  self_rhythm, self_tempo) ~ regime, data = self_octave,
  drop.unused.levels = TRUE)`: 'data' must be of a vector type, was 'NULL'
```

```
install.packages("Rarity")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
install.packages("ggfit")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
library(Rarity)
library(ggfit)
```

```
self_octave.2 <- lm(cbind(self_octave | name | regime | t) ~ subject,
                    data = self_octave, long)
```

This code fits a mixed-effects model and uses it to an object called `self_octave.2`. The response variable is `self_octave`, so the model is trying to explain differences in self-octave scores. The `cbind(self_octave | name | regime | t)` part tells the model to include the effect of `name`, the effect of `regime`, and also the interaction between them. That interaction checks whether the pattern of self-octave over time is different depending on which experimental condition a subject was in. The `~ 1 | subject` part tells the model to account for each subject, which ensures the model allows each person to have their own baseline self-octave score. The `data = self_octave, long` part tells the model where to find the data and that it is in a long format, meaning each subject appears multiple times on different rows. Using `long` saves the model from having to manually reshape the data into wide format before it can run.

`self_octave.2.long` is the name we give to the new long-format version of `self_octave.2`. `long.coef()` is the command that actually pulls out the numbers from the model. It lets us see how self-octave changes for each experimental condition, and how much each subject's self-octave changes from their own baseline. Saving this output means we can later compare, plot, or analyze those differences without rerunning the whole model.

This summary of the kind of model is not very interesting, but ggfit wants just that.

```
plot(self_octave.2, text = "Stroke")
```

```
Single term deletions

Model:
self_octave ~ (name + regime + t) | subject
       Df Sum of Sq    RSS     AIC
<NA>    0       NA 0.0000 -Inf
time:regime  6 0.017 20.5 20.8 0.0000 <0.
Warning message:
In summary.lm(.Y, ...) :
  summary.lm() produced output by "regime". You can override using the
  "print" argument.
```

regime
time
name
Stroke
Stroke
Stroke

```
32  2.14512E-26  0.12234E+02  0.45421E-17  0.Neerlia  0.Neerlia
33  2.23864E-26  0.12234E+02  0.45421E-17  0.Neerlia  0.Neerlia
34  2.23864E-26  0.12234E+02  0.45421E-17  0.Neerlia  0.Neerlia
35  2.15179E-26  0.14292E+02  5.5612E-17  0.Neerlia  0.Neerlia
36  2.15179E-26  0.14292E+02  5.5612E-17  0.Neerlia  0.Neerlia
37  2.75464E-26  0.15238E+02  1.0224E-16  0.Neerlia  0.Neerlia
38  2.71402E-26  0.15811E+02  2.0487E-16  0.Neerlia  0.Neerlia
39  2.71402E-26  0.15811E+02  2.0487E-16  0.Neerlia  0.Neerlia
40  2.2884E-26  0.17322E+02  3.3125E-16  0.Neerlia  0.Neerlia
41  2.9785E-26  1.7017E+02  2.4568E-16  0.Neerlia  0.Neerlia
42  2.9785E-26  1.7017E+02  2.4568E-16  0.Neerlia  0.Neerlia
43  2.7294E-26  2.7288E+02  4.0751E-16  0.Neerlia  0.Neerlia
44  2.2795E-26  2.9017E+02  5.5952E-16  0.Neerlia  0.Neerlia
45  2.2795E-26  2.9017E+02  5.5952E-16  0.Neerlia  0.Neerlia
46  2.4545E-26  3.1710E+02  5.9864E-16  0.Neerlia  0.Neerlia
47  2.4545E-26  3.1710E+02  5.9864E-16  0.Neerlia  0.Neerlia
48  2.4545E-26  3.1710E+02  5.9864E-16  0.Neerlia  0.Neerlia
49  2.3844E-26  7.8120E+02  6.0455E-16  0.Neerlia  0.Neerlia
50  2.1144E-26  8.4015E+02  6.0455E-16  0.Neerlia  0.Neerlia
51  2.1144E-26  8.4015E+02  6.0455E-16  0.Neerlia  0.Neerlia
52  2.1144E-26  8.4015E+02  6.0455E-16  0.Neerlia  0.Neerlia
53  2.1144E-26  8.4015E+02  6.0455E-16  0.Neerlia  0.Neerlia
54  2.1144E-26  8.4015E+02  6.0455E-16  0.Neerlia  0.Neerlia
55  2.1144E-26  8.4015E+02  6.0455E-16  0.Neerlia  0.Neerlia
```

```
195  0.621999744
196 -2.463631517
197  1.344443744
198 -2.457453814
199 -0.915630063
200 -0.178590054
201 -0.321004723
202  1.551518210
203  0.414426161
204  1.826521077
205 -0.689427508
206  2.152258959
```

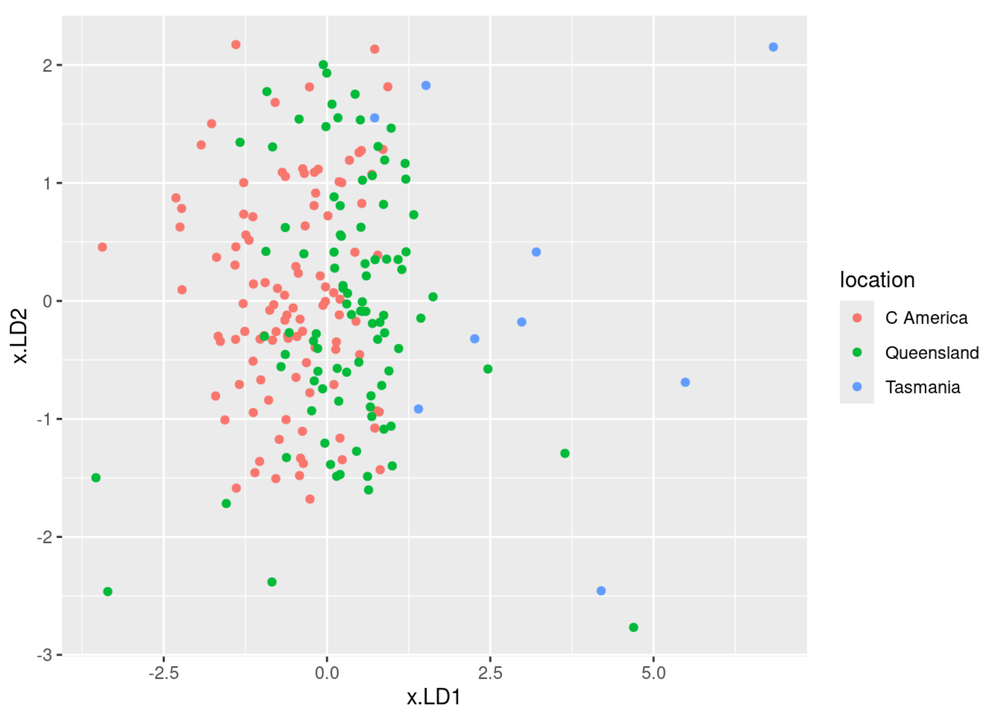
```
with(leafshape_p, table(location, class))
```

```
            class
location     C America Queensland Tasmania
  C America         79         26        0
  Queensland        29         61        2
  Tasmania           0          4        5
```

Most of the leaves have been classified correctly, but there are still a lot that are wrong. For example, the Central America and Queensland leaves are sometimes confused with each other, and only just over half of the leaves actually from Tasmania (bottom row) were actually classified as Tasmania.

In order to plot the discriminant scores against each other with the points labelled by where each leaf actually came from we have to do the following:

```
ggplot(leafshape_p, aes(x = x.LD1, y = x.LD2, colour = location)) +
  geom_point()
```



Does your plot (Question 17) tell a similar story to your misclassification table (Question 16)? Explain briefly:

Somewhat as its supported by the red points (Central America) being mostly on the left, the green ones (Queensland) being mostly in the middle, and the blue ones (Tasmania) being mostly on the right. But there is a lot of overlap, especially between Central America and Queensland (red and green), and some of the Queensland points (green) are far enough right that they look as if they came from Tasmania.

How does your plot of Question 17 support your conclusion of Question 13? Explain briefly.

In Question 13 we said that only LD1 was important, meaning that only LD1 distinguishes the locations in any meaningful way. On the plot of Question 17, the locations are only distinguished left and right (in the

LD1 direction) and not at all up and down (LD2 direction). So it is indeed true that only LD1 is important in distinguishing the locations.

Find the leaf in row 105 (of your dataframe with LD scores and posterior probabilities). Is this leaf classified correctly? Find this leaf on your graph of Question 17, and explain why its posterior probabilities make sense.

```
leafshape_p %>% slice(105) %>%
  select(contains("LD"))
```

```
        x.LD1     x.LD2
105 -3.439251 0.4560472
```

This is one of the three leftmost points on the graph, and the only one with positive LD2 score, so it is the point towards the top left, in with the ones from Central America.

```
leafshape_p %>% slice(105) %>%
  select(starts_with("posterior"))
```

```
    posterior.C America posterior.Queensland posterior.Tasmania
105           0.9590306           0.04096935       1.645016e-09
```

96% Central America, 4% Queensland, 0% Tasmania. Since the leaf is in the Central America part of the graph (on the left), it is no surprise that it is overwhelmingly likely to have been from Central America (and actually was).